MATTHEW DE LANNOY

Robotics Engineer

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PROJECTS

Cora: Collaborative Robot Platform | Mar 2025 – Present

- Lead ME/SW engineer on open-source cobots (1.0 m reach, 3 kg payload); led cross-disciplinary team of 5
- Architected distributed control system using ROS2, Ethernet TCP/IP, CAN bus motor controllers, and Python SDK
- Designed modular cobot-style joints for scalable robot architecture

MIRTE Master Laboratory Clean-up Robot (Bachelor Thesis) | Feb 2026 – Jun 2026

- Developed a complete autonomous laboratory clean-up robot integrating ROS 2, Nav2, MoveIt 2, SLAM Toolbox, and behavior-tree task execution
- Implemented RGB-D perception pipeline with point-cloud segmentation, object localization, and YOLO-based classification for autonomous grasping
- Evaluated and implemented coverage path planning algorithms while contributing to mechanical integration of the mobile manipulator and waste-sorting system

Cavolab: Automated Cable Cleaning (Robotics Minor Project) | Sept 2025 – Feb 2026

- University minor project in collaboration with Prysmian Group
- Led mechanical design of gantry system and high fidelity robot structures
- Delivered prototype validation and technical documentation for potential industry application

You vs Cobot: Chess Application | Mar 2025 – Present

- Autonomous chess system with vision-based board recognition
- CNN-based move selection showcasing full-stack robot integration

PROFESSIONAL EXPERIENCE

Data Engineer

The Green Village

June 2026 – Current

- Designed and maintained Python-based data pipelines for IoT sensor networks
- Streamed real-time sensor data to Apache Kafka for downstream processing and visualization
- Collaborated with researchers to integrate environmental sensor data into experimental smart-city applications

Teaching Assistant

TU Delft - Mechanical Engineering

Nov 2023 – Apr 2026

- Delivered laboratory instruction, assessment, and student support for Robotics courses
- Taught SolidWorks and provided mentorship in the undergraduate mechanical engineering CAD course

EDUCATION

B.Sc. Mechanical Engineering

TU Delft

Sept 2022 – June 2026

Minor in Robotics

TU Delft

Sept 2025 – Jan 2026

EXTRACURRICULAR

Workshop Committee | RSA Delft Facility management and technical workshop coordination

Hackathons

- **Aeroshield | DCSC - TU Delft**
  Ping pong ball thrower control system using open source Aeroshield
- **Underground | Samxl and RSA**
  Remote controlled duct navigating and inspection robot
- **Renewable Energy | Samxl and RSA**
  Turbine blade inspection robot for offshore wind farms
- **Boskalis | RSA**
  Autonomous seagrass planting robot for coastal restoration

TECHNICAL SKILLS

Robotics & Embedded Systems

ROS 2, Ethernet TCP/IP, CAN Bus, Raspberry Pi, RP2040, Real-time Control

Software

Python, C++, Git, Firmware Development

Design & CAD

SolidWorks, Onshape, FEA, 3D Printing, CNC Machining

Electronics

KiCAD, Microcontrollers, Motor Control, Sensor Integration

Professional

Agile/Scrum, Cross-team Communication

CERTIFICATIONS

- Certified SOLIDWORKS Expert
- CSWPA – Drawings
- CSWPA – Weldments
- CSWPA – Sheetmetal
- CSWPA – Surfacing